

Redrob Candidate Ranker

- Goal: identify the top 100 candidates for Redrob's Senior AI Engineer founding-team role.
- Built as a deterministic offline ranker: CPU-only, no network calls, no hosted LLM APIs.
- Optimized for recruiter trust: specific reasons, explicit concerns, and stable reproducibility.

What The JD Really Asks For

- Primary need: production search, retrieval, ranking, embeddings, and evaluation systems.
- Strong preference for 5-9 years, hands-on coding, product-company context, and India/Pune/Noida logistics.
- Explicit negatives: keyword-only AI exposure, pure research, services-only careers, and CV/speech-only profiles.

Scoring Architecture

- Job-fit score combines title, career narrative, trusted skills, experience band, product background, and location.
- Career evidence is weighted above raw skills to avoid candidates who only stuff AI keywords.
- Skills are trusted by proficiency, duration, endorsements, and Redrob assessment scores.

Behavioral Signals

- Availability modifier uses last active date, open-to-work flag, recruiter response rate, response time, and notice period.
- Hireability adds interview completion, offer acceptance, recruiter saves, GitHub activity, and profile verification.
- Behavior adjusts the technical score without letting a weak technical fit outrank a strong role match.

Trap And Risk Handling

- Penalizes non-technical titles with many AI skills but weak career evidence.
- Penalizes services-only histories, stale profiles, low response rates, and impossible skill-duration patterns.
- Downweights CV/speech-heavy specialization when NLP/IR/retrieval evidence is missing.

Output And Reproducibility

- Single command: `python rank.py --candidates ./candidates.jsonl --out ./submission.csv`
- Runtime on the full 100,000-candidate dataset: about 20 seconds on local CPU.
- Generated CSV passes the provided challenge validator with exactly 100 ranked rows.

Evaluation Framework

- Ranking Quality Metrics: NDCG@K, MRR, Precision@K, Recall@K
- Offline Evaluation: relevance-based ranking benchmarks and score calibration
- Online Evaluation: recruiter engagement uplift, search success rate, and candidate interaction metrics
- Future Work: A/B testing and offline-to-online correlation analysis for continuous ranking improvement